SYSTEM AND SOFTWARE DESIGN SPECIFICATION — SINGLE SOURCE OF TRUTH

BLE Smart Gate: Sensor Fusion Controller

L4-B Rework Baseline (rev 4: HIL Refinements & Expert Feedback)

Author: Architecture & Embedded Systems Consultant Group | **Date:** 17 October 2026 | **Hardware:** ProMicro nRF52840 · HC-SR04/JSN-SR04T · NC Reed Switch · 12 V Fail-Secure Strike | **Firmware:** Zephyr RTOS v4.4

Design Baseline L4-B (rev 4) Key Revisions: Incorporates zero-jitter hardware PPI/TIMER capture for ultrasonic distance, asymmetric reed debouncing (30 ms break / 150 ms make), mobile OS fast advertising intervals (20–30 ms) supporting iOS/Android background scanning, gate rebound lockout prevention, ambient temperature acoustic compensation via nRF52840 `TEMP`, and flash NVMC collision protection (`CONFIG_BT_CTLR_FLASH_GUARD=y`).

Table of Contents

1. **Introduction and Document Control** — Scope, normative conventions, baseline record
2. **Field-Test Evidence Base** — T0/T1/T2 empirical findings, multipath reality, why RSSI is dead
3. **Design Decision Log (Normative)** — The 9 core decisions, race conditions, hardware reviews
4. **User Stories** — Owner, administrator, and attacker misuser stories with acceptance criteria
5. **Pain Points and Mitigations** — Physical and UX failure modes mapped to specifications
6. **System Architecture and Block Diagram** — Subsystem topology, placement, power rails, strike polarity
7. **System Requirements** — FR-1 to FR-15 functional requirements, non-functional latency/safety goals
8. **System Behavior** — Given / When / Then execution walkthroughs and edge-case behaviors
9. **Sensor Fusion Model and Parameters** — 3-way refuse-by-default truth table, temperature compensation
10. **Gate State Machine** — 8 operational states, transitions, guards, rebound recovery logic
11. **Antenna and Placement Strategy** — Metal gate RF shielding, polarization, TX trim
12. **Security Model** — LE Secure Connections, pairing window, iOS/Android connection handling
13. **Firmware Diff Plan** — Zephyr module restructuring, PPI/TIMER driver, asymmetric debounce
14. **Static Software Design** — Architecture layers, interface contracts, data structures
15. **Dynamic Software Design** — Detailed sequence diagrams, non-blocking capture disciplines
16. **Task Allocation and Scheduling** — Execution contexts, cooperative watchdog heartbeat, stack budgets
17. **Verification Protocol (T-Series)** — Acceptance tests T-A through T-F on real gate hardware
18. **Open Items and Hardening Backlog** — Bench phase deferred items, hardware hardening roadmap
19. **Appendix A: Parameter Reference** — Master parameter table with constraints and calibration
20. **Appendix B: Requirements Traceability Matrix** — Stories to Requirements to Tests mapping
21. **Appendix C: Glossary and References** — Formal definitions and standards

1 Introduction and Document Control

This specification is the single source of truth for the redesigned BLE smart gate. It consolidates, in one normative document, the system design (user stories, pain points, requirements, behavior), the sensor-fusion architecture and its state machine, the software design (static, dynamic, task allocation), the firmware rework plan for the existing Zephyr codebase, and the verification protocol that validates the build on the physical gate. It completely supersedes the RSSI-evaluator design of earlier phases as the controlling description of product behavior.

The document is organized for three distinct engineering roles:

- **System Architects & Reviewers (Chapters 2–12):** Understand physical test evidence, design decisions, security threat modeling, and behavioral state machine contracts.

- **Firmware Engineers (Chapters 13–16, Appendix A):** Direct implementation contracts, module APIs, PPI/TIMER hardware capture, Zephyr execution contexts, and memory layouts.
- **Verification & Test Engineers (Chapters 17–18, Appendix B):** Uncut capture protocols, HIL test procedures, fault-injection criteria, and traceability.

1.1 Scope and Normative Conventions

In Scope: L4-B sensor-fusion pedestrian gate controller — BLE bonded connection as identity channel, ultrasonic distance as presence channel, reed switch as gate physical position channel, unlock-only fail-secure electric strike actuator, state machine fusing all three channels, and the Zephyr RTOS firmware reorganization.

Out of Scope: Implementation source code; companion mobile application internal UI (the background scanning and foreground service contract is normative in Chapter 12); industrial weatherized enclosure mass production (deferred to Hardening Phase 2).

Normative Keywords: `MUST` (mandatory), `SHOULD` (strong default, deviation requires formal change record), `COULD` (optional). All parameters default from firmware headers, persist in NVS, and are denoted in monospace (e.g. `usUnlockDwellMs`). Units: distances in cm, time in ms or s, power in dBm, temperature in °C.

1.2 Document Control Record

Item	Value	Revision Summary
Design Baseline	L4-B (sensor-fusion rework)	Supersedes RSSI-evaluator design based on field evidence.
Current Document Revision	Revision 4 (Approved — HIL Refinements)	Incorporates hardware PPI/TIMER capture, asymmetric debounce, iOS/Android BLE parameters, gate rebound guard, temperature compensation, and NVMC flash collision protection.
Date	17 October 2026	Baseline frozen for implementation and bench prototyping.
Evidence Base	T0/T1/T2 field captures (7 logs) + L4-A analysis	Empirical data establishing 2.4 GHz metal gate multipath reality.
Target Hardware	ProMicro nRF52840 (Cortex-M4F)	HC-SR04 (bench prototype) / JSN-SR04T (production), NC Reed Switch, 12 V Strike.

2 Field-Test Evidence Base

Every architectural decision in this document traces directly to empirical field measurements. The T-series test campaign placed an authenticated owner smartphone in a pocket across known physical positions relative to a full-size metal pedestrian gate while the ProMicro nRF52840 logged all 2.4 GHz advertisements at 1 Hz evaluation rate: T0 (stationary at gate and 3–4 m away), T1 (approach runs from 5 m), and T2 (exit runs into the house to 4–5 m).

The metal gate was the dominant physical determinant. It created extreme multipath interference: constructive and destructive interference caused the phone to read higher signal strength at 4 m away than when standing 10 cm directly in front of the gate. Consequently, **RSSI carries at most 1 bit of information ("authenticated radio present nearby: yes/no") and zero reliable distance or direction information**.

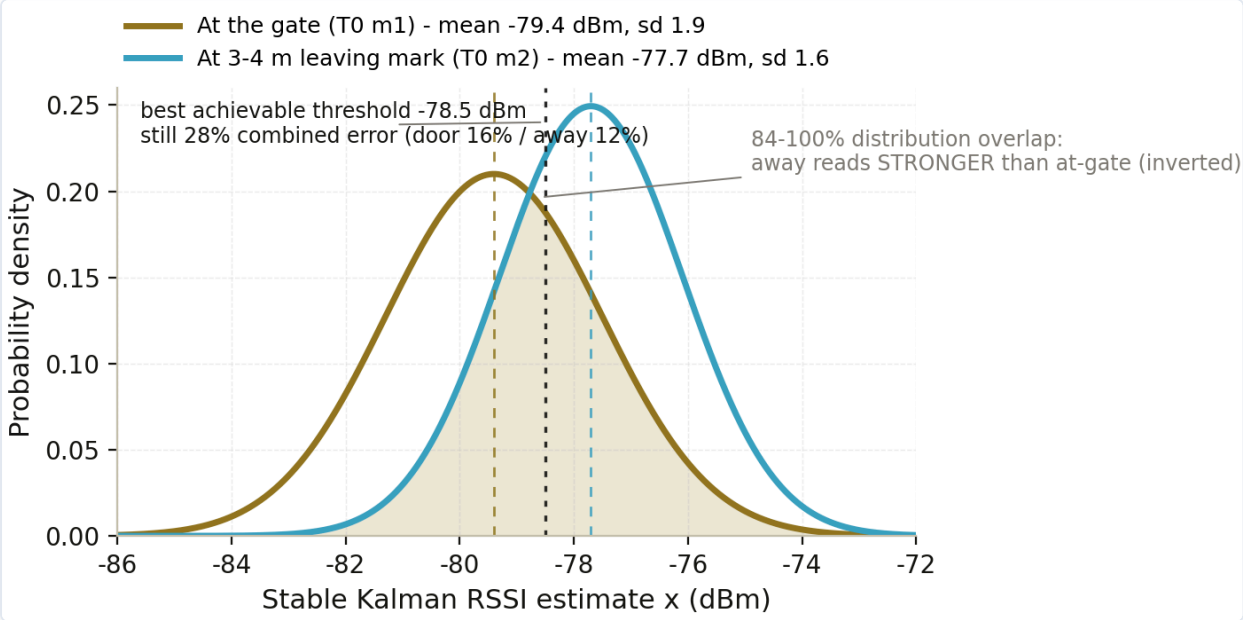


Figure 1: Stable Kalman RSSI distributions from T0 stationary windows. The 3–4 m away mark is statistically stronger (-77.7 dBm) than the gate position (-79.4 dBm); the best theoretical threshold (-78.5 dBm) still yields an unacceptable 28% classification error.

#	Measured Field Finding	Empirical Data	Design Consequence
F1	**Level thresholds are dead:** Door vs 3–4 m distributions overlap 84–100% and are statistically inverted.	Door mean -79.4 dBm (sd 1.9); away mean -77.7 dBm (sd 1.6); best threshold error = 28%.	RSSI level plays zero role in unlock decisions. BLE is strictly an identity channel (bonded link established = authenticated).
F2	**Trend slopes are dead:** Multipath noise floor exceeds walking approach slopes at 1 Hz evaluation.	Noise floor 1.79–1.89 dB/s vs walking slopes +1.0 to +1.6 dB/s. Catch rate position-independent.	Slope and trend analysis retired on unlock path. Directionality filtered by physical geometry and presence dwell.
F3	**Binary presence/absence is 100% reliable:** Link continuity survived across all scenario boundaries.	Zero false absences; zero foreign module matches. Match exact in all 7 session files.	Binary presence becomes the identity confirmation channel, while physical presence is offloaded to ultrasonic sensing.
F4	**Architecture disciplines healthy:** Sysworkq, dwell invariants, and event callbacks performed flawlessly.	Kalman normalization stable; dwell invariant held for 20 min without false level arming.	Software discipline carries over directly into `gate_sm` and sensor drivers.
F5	**Body absorption artifact:** Lifting phone out of pocket inside house added +5–6 dB over 13 s.	x: -78 to -72.8 dBm; slope peak +0.95 dB/s. Classic false-unlock trigger for level-tuned logic.	Link budget protected by vertical antenna polarization; outside gate placement shields inside spurious connections.

3 Design Decision Log (Normative)

This chapter records the binding architectural decisions approved for the L4-B baseline, updated with Revision 4 hardware-in-the-loop (HIL) engineering refinements. Any modification to these points requires a formal document revision.

#	Approved Decision	Engineering Rationale & Refinement Details
P1	**Unlock-Only Semantics:** The controller only unlocks the gate; it never commands it to lock.	The physical gate features a mechanical auto-latch that locks instantly upon closing. Eliminates the entire problem class of auto-relock timers and "inside phone keeps gate unlocked".
P2	**Ultrasonic Sensor Presence Channel:** HC-SR04 (bench prototype) / JSN-SR04T (production waterproof).	Provides unambiguous physical distance. In bench testing, dual-transducer HC-SR04 eliminates the blind zone (2 cm min). Production JSN-SR04T utilizes a 20 cm recessed mount. Acoustic speed compensated dynamically via nRF52840 internal `TEMP` sensor.
P3	**Reed Switch on Frame:** NC contact wiring; asymmetric debounce filter.	Normally Closed (NC) wiring ensures a severed wire reads OPEN (fails secure). Replaces 500 ms debounce with **30 ms on break (OPEN)** and **150 ms on make (CLOSE)** to prevent swallowing rapid human gate pushes.
P4	**Controller & Sensor Installed Outside Gate Face:** Vertically polarized.	Utilizes the steel gate leaf as an RF attenuator against indoor phone connections (Race 1 mitigation). Vertical mounting matches pocketed smartphone antenna polarization (10–20 dB link improvement).
P5	**BLE Role Dedicated to Identity & Crypto:** LE Secure Connections, bonding, authenticated passkey.	Eliminates scan-mode UUID guessing. Link encryption guarantees identity; presence and position decide actuation.
P6	**Exit Connection Drop:** If connection forms while reed is OPEN, drop link for `reconnectHoldoffMs` (10 s).	Indicates user is walking out from inside; prevents activating ultrasonic or strike while gate is already ajar.
P7	**De-arm on Reed Break:** Relay energizes strike; on reed OPEN, MCU immediately pulls relay IN LOW to re-arm.	De-energizes strike before user completes transit; mechanical auto-latch catches and locks the moment the gate closes.
P8	**Actuator Duty Cycle Protection:** `failedAttemptTimeoutMs` (10 s) followed by equal cooldown.	Protects electric strike solenoid from continuous energization burnout; bounds duty cycle to ≤ 50%.
P9	**Rate Limiting & Lockout:** 3 consecutive failed attempts trigger 15-minute `BLOCKED` state.	Persisted in NVS with CRC32 so power cycling cannot reset lockout. Manual mechanical key operates normally.

3.1 Race Condition Mitigations

Race ID	Hazard Scenario	Approved Resolution (Normative)
R1	**Decoupled Identity:** Owner's phone connects from inside while stranger stands at gate.	Layered defense: (a) US-gated advertising window — radio remains silent until body arrives; (b) gate acts as RF shield; (c) `txPowerDbm` trimmed to approach zone. Tested by T-F.
R2	**Exit Lock-Up:** User exits, closes gate, stands outside; phone reconnects and unlocks gate behind them.	Deterministic guard: Reed OPEN-to-CLOSED transition triggers `exitGraceMs` (60 s) refusing unlock regardless of BLE. **Rev 4 Guard:** If close occurs within 3 s of an external unlock with continuous US presence, treated as gate rebound, not exit grace.
R3	**Position vs Lock State:** Reed detects frame closure, not bolt throw.	Closed by mechanical auto-latch: closed position guarantees physical latching. Latch-monitoring strike tracked for Hardening Phase 2.
R4	**Walk-by Unlock:** Owner walks past gate along sidewalk; phone connects and gate unlocks.	Calibrated `maxDetectableDistance` + N-of-M debouncing + 2.0 s dwell filter. Passing within beam for <2 s aborts unlock.

Race ID	Hazard Scenario	Approved Resolution (Normative)
R5	**Flash NVMC Stalls:** NVS page erase stalls CPU, causing BLE supervision timeout.	Enabled `CONFIG_BT_CTLR_FLASH_GUARD=y`. Transient attempt counters kept in RAM; NVS writes occur only on permanent lockout (`BLOCKED`) or admin config save.

3.2 Hardware and Firmware Decisions

Subsystem	Approved Design Baseline	Revision 4 Technical Note
Ultrasonic Timing	Hardware PPI + 32-bit TIMER capture.	Replaces software GPIO ISR. Zero-jitter (<1 µs) capture immune to BLE radio priority interrupts.
Ultrasonic Power	5 V from VBUS; bidirectional level shifting.	Ensures V_IH ≥ 3.5 V on TRIG input and safe V_OH ≤ 3.3 V on nRF52840 ECHO pin.
Strike Transient Protection	Antiparallel 1N4007 + 15 V TVS across coil; MCU Schottky + 1000 µF buffer.	Mandatory hardware hardening specification. Deferred to Bench Phase 2 due to physical terminal accessibility.
BLE Advertising	Fast connectable advertising (20–30 ms) during US window.	Supports both Android (foreground service) and iOS CoreBluetooth background scanning duty cycles.
Watchdog Timer	Hardware WDT (8 s) with cooperative heartbeat.	`main.c` feeds WDT only if `sysworkq` heartbeat counter has incremented, catching task queue deadlocks.

4 User Stories

System requirements derive from testable user stories covering owners, administrators, and deliberate misusers/attackers.

ID	Actor & User Story	Acceptance Criteria & Verification
US-1	As the approaching owner with phone in pocket, I want the gate to unlock hands-free.	Unlocks within 2.5–3.5 s of arrival; zero manual phone interaction; state trace IDLE → LINK → UNLOCK_HOLD → ARM. (T-B)
US-2	As the exiting owner, I want to leave and close the gate without it unlocking behind me.	Connection dropped on open; on close, 60 s `exitGrace` enforces locked state; zero false unlocks. (T-C)
US-3	As the owner, I want the mechanical key to always work under any failure, block, or blackout.	Fail-secure strike (de-energized = locked); lock cylinder operates independently of electronics. (T-E)
US-4	As a blocked owner, I want clear status feedback without permanent lockout.	GATT status characteristic reports BLOCKED with remaining minutes; LED pulses triple blink; resets after 15 min. (T-E)
US-5	As the administrator, I want to calibrate parameters in NVS without recompiling.	Parameters readable/writable via console and GATT; CRC integrity check; runtime limits enforced. (T-A, T-F)
US-6	As the owner resting inside my home, I want my nearby phone not to cause spurious unlocks.	US-gated window: radio stays silent when gate is unoccupied; gate leaf provides RF shielding. (T-F)
US-7	As the owner during a power failure, I want the gate to remain secure.	Fail-secure strike remains locked; on power restoration, boot state defaults to locked; block state restored from NVS. (T-E)
US-8	As a passer-by or unbonded visitor, the gate must never open for me.	Unbonded link rejected; presence alone can never fire relay (fusion truth table). (T-B, T-F)
US-9	As the maintainer, I want structured telemetry and state observability.	Deferred LOG emissions on all transitions; non-blocking CDC ACM console status snapshot. (All tests)
M-1	Attacker presents unbonded phone at gate.	Security handshake fails; link terminated within `smpFailTimeoutMs` (2.5 s); relay unaffected.
M-2	Accomplice at gate while owner's phone is inside house (Race 1).	Mitigated by RF shielding and power trim; residual risk accepted for residential grade. (Ch. 12)
M-3	Gate-shaker: repeated forced triggers causing continuous timeout.	Strike protected by 10 s max energization; after 3 failures, 15 min lockout engaged.
M-4	Blinder: covering ultrasonic sensor with tape/obstacle.	Continuous blind echo or no-echo triggers `US_FAULT` after 10 s; auto-unlock refused; manual key works.
M-5	Rogue phone attempts pairing during maintenance.	Pairing responder disabled by default; opens only via physical button/console for ≤ 60 s with SC passkey.

5 Pain Points and Mitigations

Pain Point	Root Cause	Normative Mitigation in L4-B (rev 4)
Multipath Inversion	2.4 GHz RF reflects off steel gate; away signal exceeds gate signal.	RSSI completely removed from decision path. BLE demoted to binary identity; presence measured by ultrasonic.
Hands-Free Lag	BLE discovery and connection setup takes 1–2 s.	Parallel critical path: fast 2-of-3 presence opens advertising in 0.3 s; phone connects concurrently with 2.0 s dwell. Total 2.5–3.5 s.
Actuator Burnout	Solenoids overheat if energized continuously during jam.	Hardware timer limits fire time to `failedAttemptTimeoutMs` (10 s) with mandatory equal cooldown period.
Gate Rebound Lockout	Light push causes door to bounce off rubber seal into latch.	State machine checks if US presence is still continuously active; cancels `EXIT_GRACE` if re-closure occurs within 3 s.
Acoustic Drift	Outdoor temperature changes alter speed of sound by >10%.	Internal nRF52840 `TEMP` peripheral measures die/ambient temp and adjusts distance calculation dynamically.
BLE Drop on Flash Erase	NVMC page erase halts CPU instruction bus for 85 ms.	Enabled `CONFIG_BT_CTLR_FLASH_GUARD=y` ; flash writes restricted to lockout entry and config saves.
BLE Advertising Misses	Smartphone OS throttles background scanning when screen is off.	Gate advertises at aggressive 20–30 ms intervals during the active presence window, catching mobile scan windows rapidly.

6 System Architecture and Block Diagram

The system consists of three sensory inputs, one actuator, and the ProMicro nRF52840 controller running Zephyr RTOS. The board is mounted on the outside face of the steel pedestrian gate, vertically polarized, co-located with the ultrasonic sensor. The NC reed switch is installed on the gate frame with its actuator magnet on the moving leaf. The relay and 12 V fail-secure electric strike are housed in the gate post.

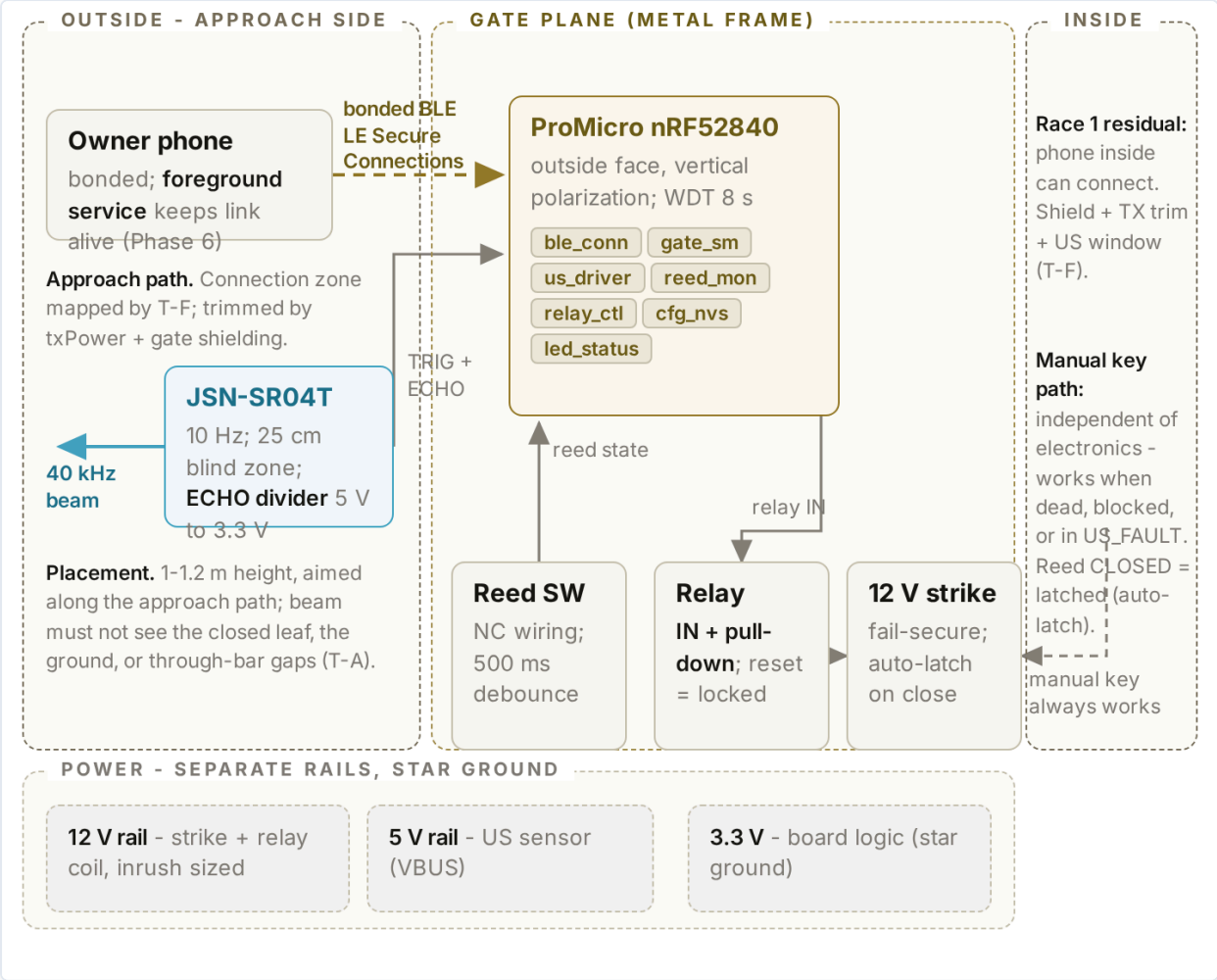


Figure 2: System block diagram showing power isolation, sensing channels, controller, and actuator zones.

6.1 Actuator Truth Table & Electrical Safety

System Operating State	Relay IN Control Pin	Strike Solenoid	Physical Gate State
Boot / Reset / Brownout / Power Loss	LOW (External 10 kΩ Pull-Down)	De-energized (Locked)	Securely Locked (Fail-Secure)
UNLOCK_HOLD (All 3 Criteria True)	HIGH (MCU GPIO Driven)	Energized (Unlocked)	Unlatched / Free to Push
ARM (Transit in Progress: Reed OPEN)	LOW (MCU De-energizes)	De-energized (Re-armed)	Ajar; Auto-latches on close
COOLDOWN / BLOCKED / US_FAULT / IDLE	LOW	De-energized (Locked)	Securely Locked

Normative Placement & Wiring Rules:

- **Ultrasonic Transducer:** Mounted at 1.0–1.2 m height, aimed along the exterior approach path. Bench prototype uses HC-SR04 (dual transducer, 2 cm min range) set back from gate. Production JSN-SR04T requires a 20 cm recessed mount.

- **Reed Switch:** NC wiring (broken wire reads OPEN = refuse to unlock). Mounted on 15 mm non-ferrous standoffs to prevent steel gate magnetic shunting. Asymmetric debounce: 30 ms break / 150 ms make.
- **Actuator Transient Suppression (Hardware Hardening Phase 2):** 1N4007 fast freewheeling diode + 15 V bidirectional TVS diode installed directly across strike terminals; 1000 µF bulk electrolytic capacitor + SS34 Schottky diode on MCU input supply rail.

7 System Requirements

7.1 Functional Requirements

ID	Requirement Statement	Traced Stories & Tests
FR-1	The gate controller SHALL accept BLE connections only from bonded peers utilizing LE Secure Connections with link encryption.	US-1, US-8 / T-F, T-E
FR-2	The gate controller SHALL advertise only when ultrasonic presence is confirmed (2-of-3 at 10 Hz) with an advertising interval of 20–30 ms, and no guard is active.	US-6, M-2 / T-B, T-F
FR-3	Relay IN SHALL transition HIGH only when all three hold simultaneously: bonded encrypted connection, reed CLOSED, and ultrasonic presence dwell ≥ `usUnlockDwellMs` (2.0 s).	US-1 / T-B
FR-4	The controller SHALL operate in unlock-only mode; relay IN SHALL transition LOW immediately upon `REED_OPEN` to re-arm the strike.	US-2 / T-C
FR-5	A connection established while the reed switch is OPEN SHALL be dropped within 100 ms for `reconnectHoldoffMs` (10 s) without ultrasonic firing.	US-2, US-6 / T-C
FR-6	A reed OPEN-to-CLOSED transition SHALL initiate `exitGraceMs` (60 s) refusing unlock, unless identified as an immediate entry rebound (<3 s with US presence).	US-2 / T-C
FR-7	If reed OPEN is not detected within `failedAttemptTimeoutMs` (10 s) during `UNLOCK_HOLD`, IN SHALL go LOW, followed by `strikeCooldownMs`.	M-3 / T-E
FR-8	Accumulating `failedAttemptCounter` (3) consecutive timeouts without success SHALL engage `BLOCKED` state for `blockDurationMin` (15 min).	US-4, M-3 / T-E
FR-9	Lockout state SHALL survive power reset via NVS with CRC32. Flash write operations SHALL be guarded by `CONFIG_BT_CTLR_FLASH_GUARD=y`.	US-4, US-7 / T-E
FR-10	A custom GATT Status Characteristic SHALL expose system state (NORMAL / BLOCKED / COOLDOWN / US_FAULT) and mirror state to status LEDs.	US-4 / T-E
FR-11	Absence of valid echo for `usFaultThresholdMs` (10 s) during active sampling SHALL raise `US_FAULT`, refusing auto-unlock until 30 s continuous valid echo.	M-4 / T-E
FR-12	A hardware Watchdog Timer (8 s) SHALL be fed cooperatively by the main loop predicated on active `sysworkq` progress ticks.	US-7 / T-E
FR-13	The mechanical lock cylinder SHALL operate independently of the electric strike and electronic controller in all states.	US-3 / T-E
FR-14	Every state transition, guard evaluation, and fault event SHALL emit a deferred log line; console SHALL provide an interactive status command.	US-9 / All Tests
FR-15	Pairing mode SHALL be disabled by default and accessible only within an administrator-initiated window ≤ 60 s using LE Secure Connections.	US-5, M-5 / T-E

7.2 Non-Functional Requirements

- **Unlock Latency:** 2.5–3.5 s end-to-end from physical arrival at gate plane. Stretch target: 2.2 s with fast smartphone.
- **Fail-Secure Integrity:** 100% of single-point component failures (power loss, cut wires, firmware hang) result in locked state.
- **Strike Safety:** Duty cycle strictly bounded ≤ 50%; energization duration never exceeds rated coil continuous rating.
- **Timing Jitter:** Ultrasonic echo pulse capture jitter < 1.0 µs achieved via hardware PPI/TIMER.

8 System Behavior

Operational Scenario	Behavioral Specification (Given / When / Then)
Normal Hands-Free Arrival	**Given** gate is locked and idle. **When** authenticated owner approaches: fast 2-of-3 presence opens BLE advertising (0.3 s); phone connects and encrypts link (1–2 s) while 2.0 s dwell accumulates; **Then** all 3 agree, relay IN goes HIGH, strike unlatches; user pushes gate (reed OPEN), IN goes LOW; gate swings shut, auto-latches, enters `EXIT_GRACE` (60 s).
Gate Rebound on Entry	**Given** gate unlatches on arrival. **When** user pushes lightly and gate rebounds off weatherstripping into latch within 3 s while user is still outside; **Then** system identifies active US presence, bypasses `EXIT_GRACE`, and resets to `IDLE`, preventing external lockout.
Exit from House (Race 2)	**Given** owner walks out from house. **When** gate opens, connection drops; on close, `exitGraceMs` (60 s) activates; **Then** if phone connects while standing outside talking, unlock is hard-refused; system enters `IDLE` after grace expires with no presence.
Sidewalk Walk-By (Race 4)	**Given** owner walks past gate along sidewalk at 2 m distance. **When** US beam detects transient presence; **Then** advertising may briefly start, but 2.0 s dwell never accumulates; advertising terminates when presence lost; zero unlocks fire.
Mid-Attempt BLE Drop	**Given** state is `UNLOCK_HOLD` (strike energized). **When** phone drops connection (body RF shielding); **Then** unlock hold continues bounded by 10 s timeout as long as US presence holds; aborts immediately only if US presence is also lost.
Strike Jam / Timeout	**Given** gate is mechanically jammed. **When** 10 s expires without reed OPEN; **Then** IN drops LOW, 10 s cooldown begins, attempts counter increments; on 3rd failure, `BLOCKED` (15 min) engages, written to NVS with CRC.
Ultrasonic Obstruction	**Given** sensor is iced, blind, or covered. **When** no-echo persists for 10 s during sampling; **Then** `US_FAULT` state engaged; auto-unlock disabled; status characteristic notifies fault; manual key operates freely.

9 Sensor Fusion Model and Parameters

The sensor fusion model allocates each decision parameter to a single authoritative physical channel:

- **Identity & Cryptographic Authorization:** BLE Bonded Encrypted Connection.
- **Physical Presence & Intention:** Ultrasonic Distance Stream with Sustained Dwell.
- **Gate Closure & Lock Engagement:** NC Magnetic Reed Switch on Frame.

9.1 Acoustic Speed Temperature Compensation

Ultrasonic distance conversion must account for outdoor temperature variations:

```
c_air(T) = 331.3 * sqrt(1 + T_celsius / 273.15) ≈ 331.3 + (0.606 * T_celsius) m/s
Distance (cm) = (Echo_Time_us * (331.3 + 0.606 * T_nrf_die)) / 20000.0
```

Ambient temperature is sampled from the internal nRF52840 `TEMP` peripheral (±0.5°C accuracy) at 0.1 Hz on `sysworkq`, eliminating up to ±16.5 cm of acoustic drift across -10°C to +50°C.

9.2 Fusion Truth Table (Refuse-by-Default)

Bonded Link	Reed Switch	US Presence Stream	Active Guards	Fusion Decision & System Action
No	Any	Any	None	**LOCKED.** Advertising window opened only if US presence confirmed.
Yes	OPEN	Any	None	**EXIT DETECTED.** Drop connection for `reconnectHoldoffMs` (10 s); no strike fire.
Yes	CLOSED	No / Lost	None	**LOCKED.** Window open; waiting for presence dwell.
Yes	CLOSED	Yes (Dwell < 2.0 s)	None	**ARMING.** Dwell accumulating on US stream; strike de-energized.
Yes	CLOSED	Yes (Dwell ≥ 2.0 s)	None	**UNLOCK.** Relay IN → HIGH; start 10 s timeout.
Yes	CLOSED	Yes (Dwell ≥ 2.0 s)	`exitGrace` / `reconnectHoldoff`	**REFUSED.** Guards override 3-way AND; strike remains de-energized.
Any	Any	Any	`BLOCKED`	**REFUSED.** Lockout active; status characteristic read-only.

10 Gate State Machine

The state machine implemented in `gate_sm` is the normative core of the firmware. It is the sole component permitted to command `relay_ctl`.

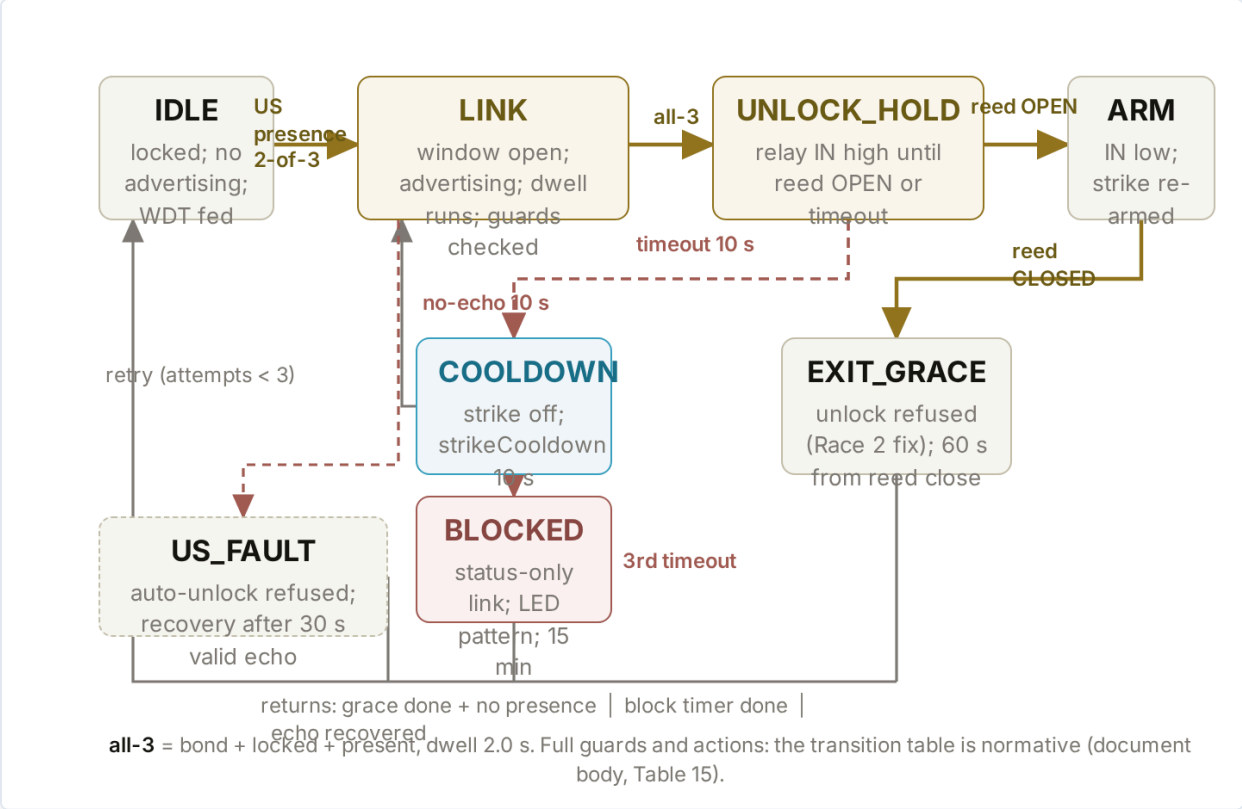


Figure 3: Gate finite state machine diagram. Solid brass lines indicate happy-path entry cycle; dashed red lines indicate fault, timeout, and lockout branches.

10.1 Normative State Transition Table

Current State	Event [Guard Condition]	State Transition Action	Next State
`IDLE`	US presence confirmed (2-of-3) [Not blocked, no grace]	Start fast advertising (20–30 ms); start dwell timer	`LINK`
`LINK`	All-3 true (Bonded + Reed CLOSED + Dwell ≥ 2s)	Relay IN → HIGH; start 10 s attempt timeout	`UNLOCK_HOLD`
`LINK`	US presence lost (N-of-M lost)	Stop advertising; reset dwell timer	`IDLE`
`LINK`	No-echo duration ≥ `usFaultThresholdMs` (10 s)	Log fault; stop advertising	`US_FAULT`
`UNLOCK_HOLD`	Reed OPEN (Gate physically opening)	Relay IN → LOW (re-arm strike); stop attempt timeout	`ARM`
`UNLOCK_HOLD`	Timeout (10 s) [Reed remained CLOSED]	Relay IN → LOW; attempts++; start 10 s cooldown	`COOLDOWN`
`UNLOCK_HOLD`	BLE dropped [US presence continuously valid]	Continue energization bounded by 10 s timeout	`UNLOCK_HOLD`
`UNLOCK_HOLD`	BLE dropped AND US presence lost	Abort: Relay IN → LOW; attempts++; start cooldown	`COOLDOWN`
`ARM`	Reed CLOSED [US present AND (now - t_unlock < 3000ms)]	Log entry rebound; bypass exit grace; reset dwell	`IDLE`
`ARM`	Reed CLOSED [Otherwise / normal transit complete]	Start `exitGraceMs` (60 s); stop advertising	`EXIT_GRACE`

Current State	Event [Guard Condition]	State Transition Action	Next State
`EXIT_GRACE`	Grace timer done (60 s) AND no presence	Clear grace guard	`IDLE`
`COOLDOWN`	Cooldown timer done [attempts < `failedAttemptCounter`]	Resume sensing assessment	`LINK`
`COOLDOWN`	Cooldown timer done [attempts ≥ `failedAttemptCounter`]	Commit lockout to NVS (CRC); trigger LED pattern	`BLOCKED`
`BLOCKED`	Block timer done (15 min)	Clear block in NVS; attempts = 0	`IDLE`
`BLOCKED`	Connection request from bonded phone	Accept status-only link; notify remaining minutes	`BLOCKED`
`US_FAULT`	Valid echo received continuously for 30 s	Log fault recovery	`IDLE`
`Any`	BLE connection formed while reed is OPEN	Drop connection; enforce `reconnectHoldoffMs`	`Same`

11 Antenna and Placement Strategy

At 2.4 GHz, small PCB trace antennas adjacent to large steel structures are essentially omnidirectional; directivity cannot be created by antenna trace geometry. Instead, physical spatial boundaries are established via three mechanisms:

- **Metal Gate Ground Plane Shielding:** Mounting the controller exclusively on the exterior metal face forces RF paths toward the interior house to pass through the steel gate leaf, providing 15–25 dB attenuation.
- **Co-Polarization:** The nRF52840 PCB is mounted strictly vertical, aligning with vertical pocketed smartphone orientation. This counteracts the 10–20 dB torso attenuation observed in field finding F5.
- **Output Power Trim (`txPowerDbm`):** Configured via NVS (default 0 dBm, adjustable down to -8 dBm) based on T-F field grid mapping to restrict the RF link budget strictly to the approach walkway.

12 Security Model

12.1 BLE Role Assignment & Mobile OS Compatibility

Dimension	Approved Baseline (Gate Peripheral / Phone Central)	Prior Assignment (Gate Scanner / Phone Advertiser)
Mobile OS Compatibility	Native CBCentralManager (iOS) and BluetoothLeScanner (Android) background connection support.	Severely throttled background advertising when phone is locked; silent discovery failures.
Advertising Latency	Fast 20–30 ms advertising interval during US window aligns with iOS low-duty background scan windows.	Phone advertised at 1 Hz; 3–4 s latency floor.
Owner Privacy	Phone transmits nothing when away; gate uses rotating resolvable private addresses (RPA).	Phone beaconed static UUID 24/7, inviting third-party tracking.

12.2 Threat Model and Residual Risk Matrix

Threat Class	Attack Mechanism	Defensive Countermeasure	Residual Risk Status
Unbonded Interloper	Rogue smartphone attempts connection at gate.	LE Secure Connections; link encryption required; unbonded peers rejected.	Zero risk.
Relay Attack	Accomplice relays BLE packets from owner inside house.	US-gated advertising window requires human at gate; short dwell window.	Accepted residual risk (residential grade).
Sensor Blind DoS	Attacker coats ultrasonic sensor in acoustic foam or tape.	`US_FAULT` state engaged after 10 s; auto-unlock disabled; mechanical key functional.	Denial of service accepted; security maintained (fails secure).
Rogue Pairing	Attacker attempts pairing while admin is nearby.	Pairing responder OFF by default; opened via physical button for ≤ 60 s with SC passkey.	Requires physical administrative compromise.

13 Firmware Diff Plan (No Source Code)

Module Path	Diff Action	Architectural Content & Zephyr Rationale
`src/ble/scan.c, scan.h`	**DELETE**	Obsolete. Scanner role retired in favor of connectable peripheral advertising.
`src/ble/proximity.c, proximity.h`	**DELETE**	RSSI evaluator, Kalman filter, and trend algorithms deleted. Dwell invariant moved to `gate_sm`.
`src/ble/ble_conn.c, ble_conn.h`	**ADD**	Manages connectable advertising window (20–30 ms fast interval), LE Secure Connections, bonding, and GATT Status service.
`src/HAL/us_driver.c, us_driver.h`	**ADD**	Non-blocking ultrasonic driver using **hardware PPI + 32-bit TIMER capture**. Ambient temperature compensation via `TEMP` peripheral.
`src/HAL/reed_mon.c, reed_mon.h`	**ADD**	GPIO interrupt with asymmetric debouncing: 30 ms break (OPEN) / 150 ms make (CLOSE).
`src/app/gate_sm.c, gate_sm.h`	**ADD**	Core finite state machine (Chapter 10). Sole owner of relay actuation and guard tracking.
`src/nvs/cfg_nvs.c, cfg_nvs.h`	**ADD**	NVS persistence with CRC32. Writes only on lockout entry or config change. Guarded by `CONFIG_BT_CTLR_FLASH_GUARD=y`.
`src/main.c`	**REWORK**	Sequential init, cooperative WDT heartbeat loop, CDC ACM shell status command.
`prj.conf`	**REVISE**	Enable `CONFIG_BT_PERIPHERAL=y`, `CONFIG_BT_SMP=y`, `CONFIG_BT_CTLR_FLASH_GUARD=y`, `CONFIG_NRFX_TIMER1=y`, `CONFIG_NRFX_PPI=y`.

14 Static Software Design

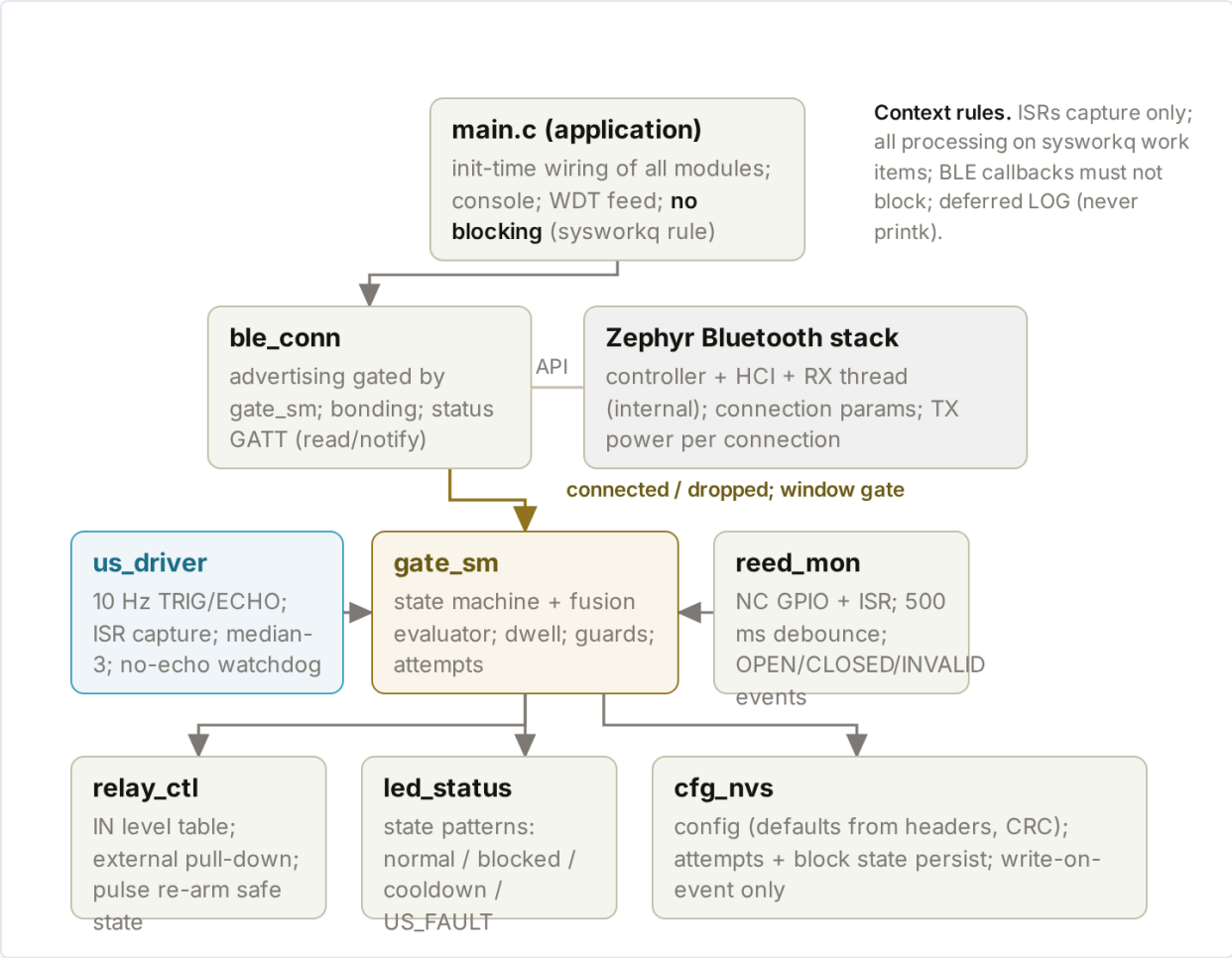


Figure 4: Static design module layering graph. Arrows indicate permitted calling dependencies; gate_sm is the sole owner of actuation.

14.1 Module Interface Contracts

Module	Key Public API Functions	Execution Context & Thread Rules
`us_driver`	`us_driver_start()`, `us_driver_read(&sample)`	PPI/TIMER captures in HW; 100 ms trigger work item runs on `sysworkq`.
`reed_mon`	`reed_mon_init(callback)`	GPIO ISR queues debounced work item to `sysworkq`.
`gate_sm`	`gate_sm_on_us()`, `gate_sm_on_reed()`, `gate_sm_on_ble()`	Runs strictly on `sysworkq`. Sole caller of `relay_ctl`.
`ble_conn`	`ble_conn_set_window(open)`, `ble_status_notify(val)`	Zephyr BT callbacks offload events to `sysworkq`.
`relay_ctl`	`relay_arm()`, `relay_energize()`	Direct GPIO write on `sysworkq`. External pull-down safe state.
`cfg_nvs`	`cfg_param_get()`, `cfg_param_set()`, `cfg_block_commit()`	Guarded flash writes on `sysworkq`. Write-on-lockout only.

14.2 Shared Data Structures

```
enum gate_state {
    STATE_IDLE, STATE_LINK, STATE_UNLOCK_HOLD, STATE_ARM,
    STATE_EXIT_GRACE, STATE_COOLDOWN, STATE_BLOCKED, STATE_US_FAULT
```

```
};

enum gate_event {
    EVT_US_PRESENCE, EVT_US_LOST, EVT_US_FAULT, EVT_US_RECOVER,
    EVT_REED_OPEN, EVT_REED_CLOSED, EVT_REED_REBOUND,
    EVT_BLE_CONNECTED, EVT_BLE_DROPPED,
    EVT_T_DWELL_DONE, EVT_T_TIMEOUT, EVT_T_COOLDOWN, EVT_T_GRACE, EVT_T_BLOCK
};

struct cfg_params {
    uint16_t maxDetectableDistance_cm;  // Default: 150 cm
    uint16_t usHysteresis_cm;            // Default: 10 cm
    uint16_t usUnlockDwellMs;            // Default: 2000 ms
    uint8_t  usDebounceN_M;              // Default: 3-of-5 (strict)
    uint32_t exitGraceMs;                // Default: 60000 ms
    uint32_t reconnectHoldoffMs;         // Default: 10000 ms
    uint32_t failedAttemptTimeoutMs;     // Default: 10000 ms
    uint32_t strikeCooldownMs;           // Default: 10000 ms
    uint8_t  failedAttemptCounter;       // Default: 3
    uint16_t blockDurationMin;           // Default: 15 min
    int8_t   txPowerDbm;                 // Default: 0 dBm
    uint8_t  wdtTimeoutS;                // Default: 8 s
    uint16_t usFaultThresholdMs;         // Default: 10000 ms
    uint16_t reedDebounceOpenMs;         // Default: 30 ms (break)
    uint16_t reedDebounceCloseMs;        // Default: 150 ms (make)
    uint16_t bleAdvFastIntervalMinMs;    // Default: 20 ms
    uint16_t bleAdvFastIntervalMaxMs;    // Default: 30 ms
    bool     tempCompEnabled;            // Default: true
    uint32_t crc32;
};
```

15 Dynamic Software Design

The sequence diagrams below establish the normative interaction patterns across Zephyr execution contexts.

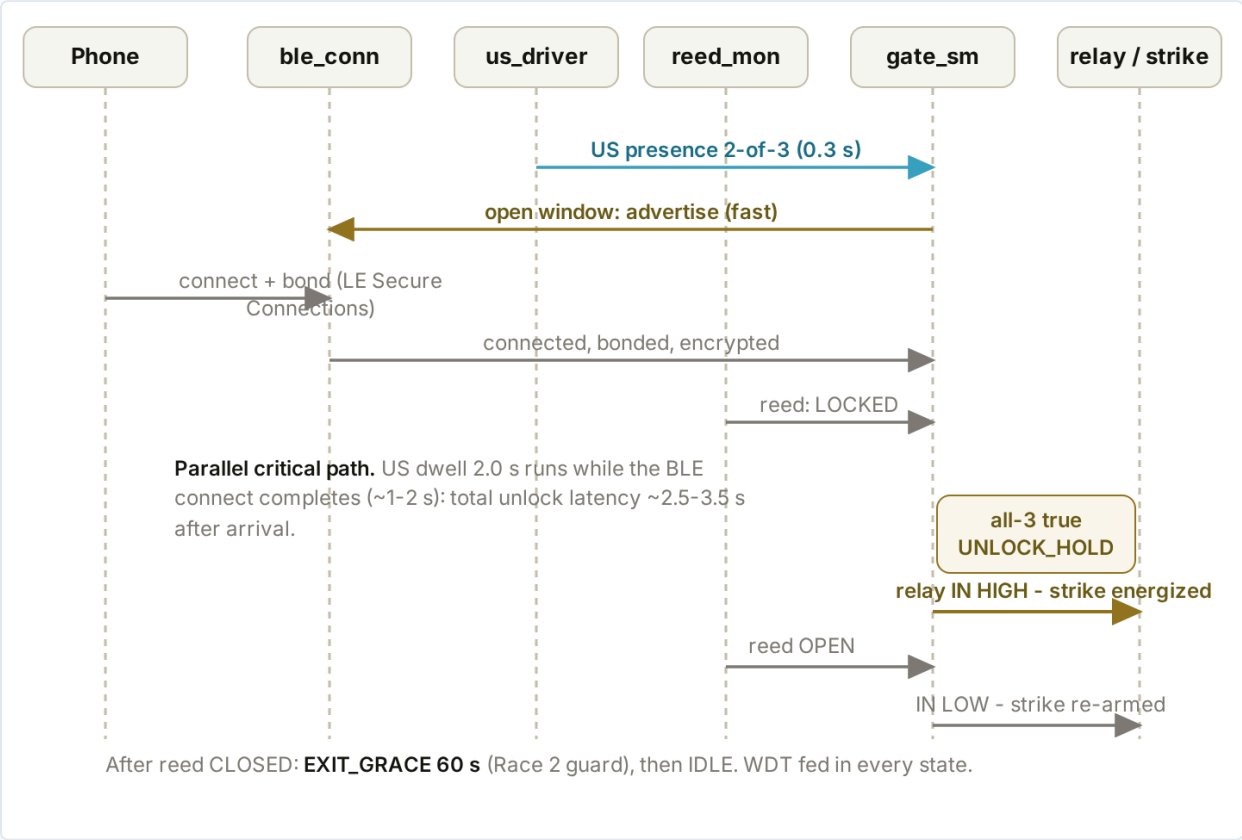


Figure 5: Sequence diagram for normal hands-free arrival. BLE connection and ultrasonic dwell progress concurrently.

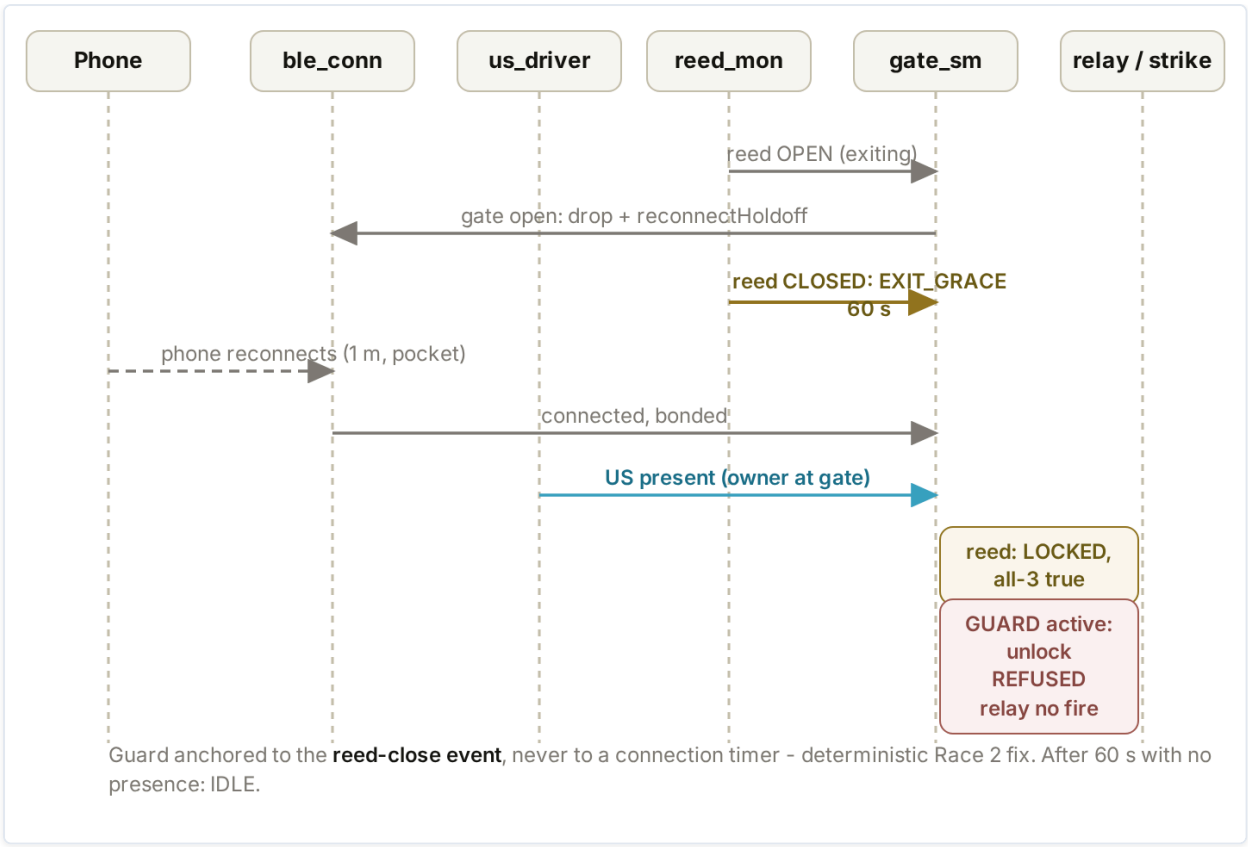


Figure 6: Sequence diagram for owner exit and closure. Reed close event anchors EXIT_GRACE; rebound guard active.

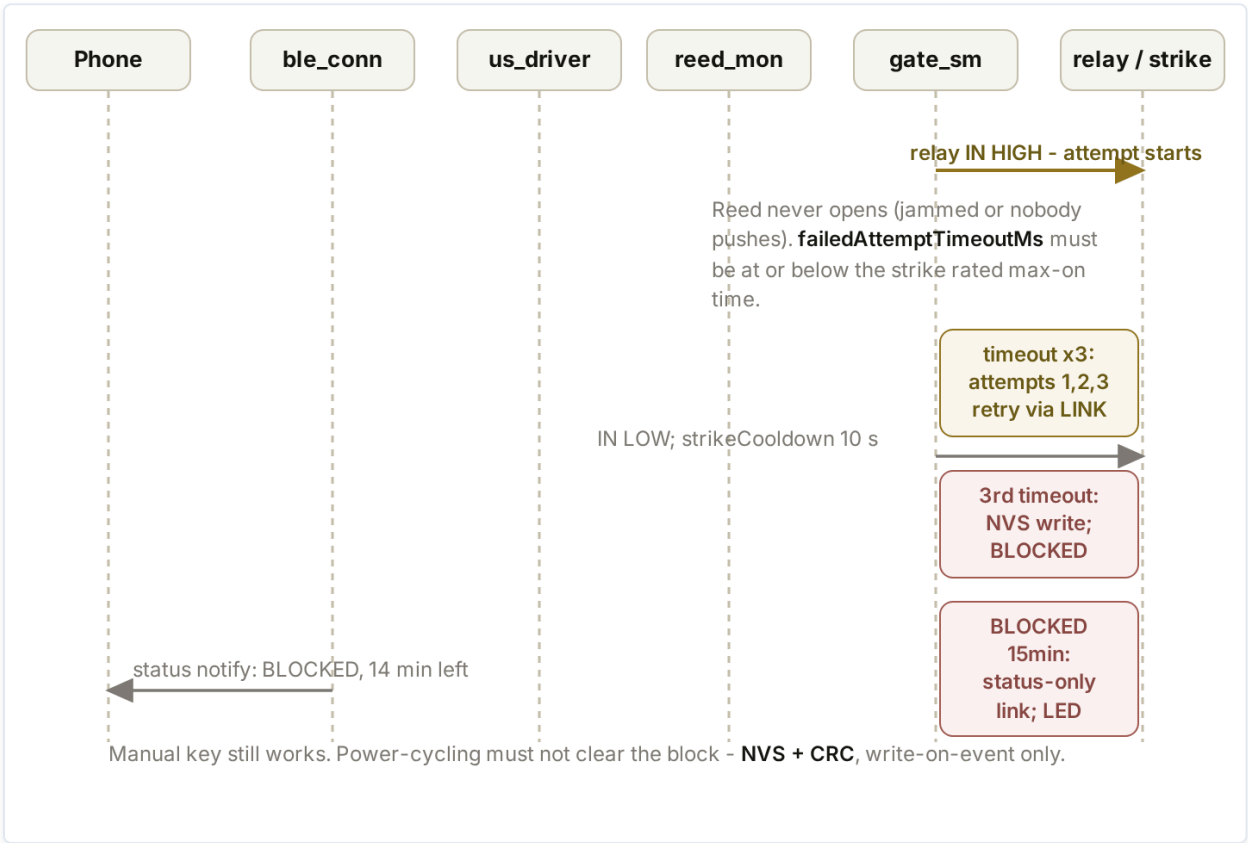


Figure 7: Sequence diagram for failed attempt timeout, strike cooldown, and NVS lockout persistence.

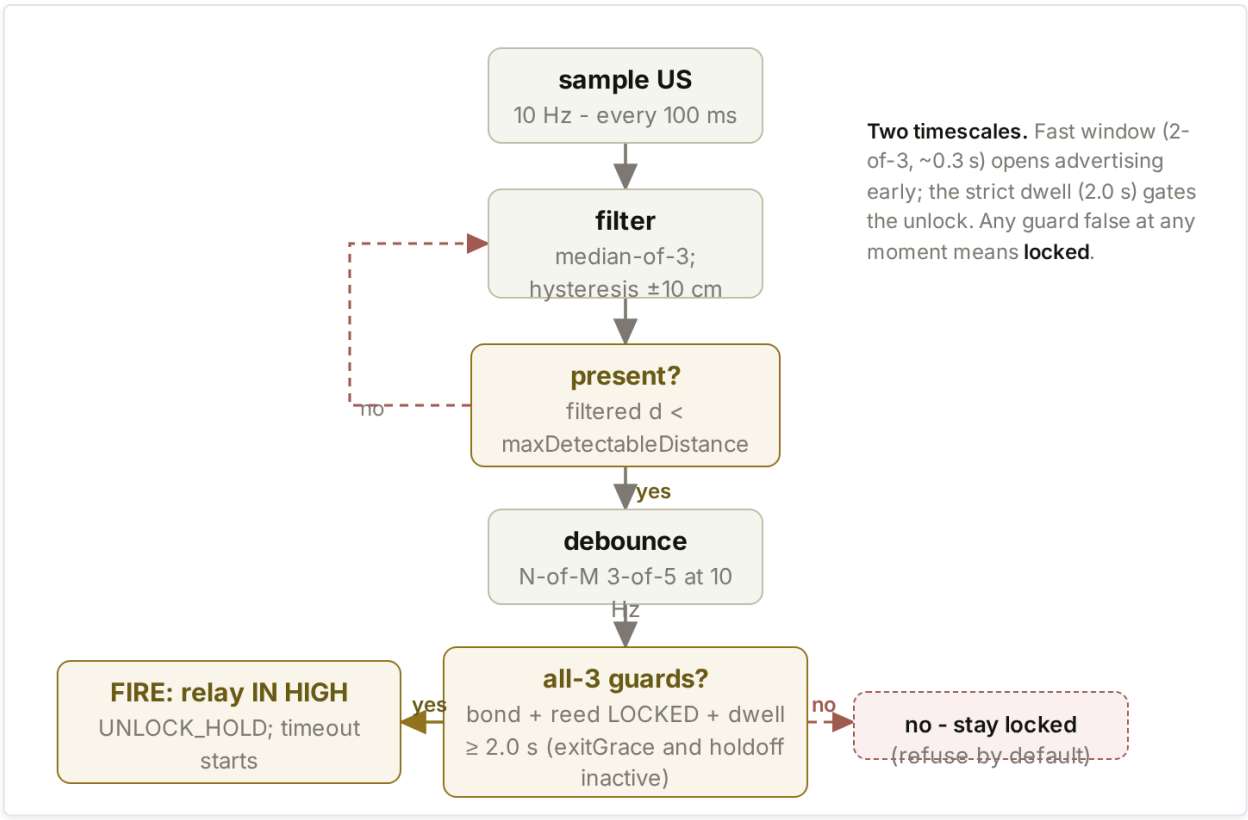


Figure 8: Decision evaluation flowchart demonstrating strict refuse-by-default logic.

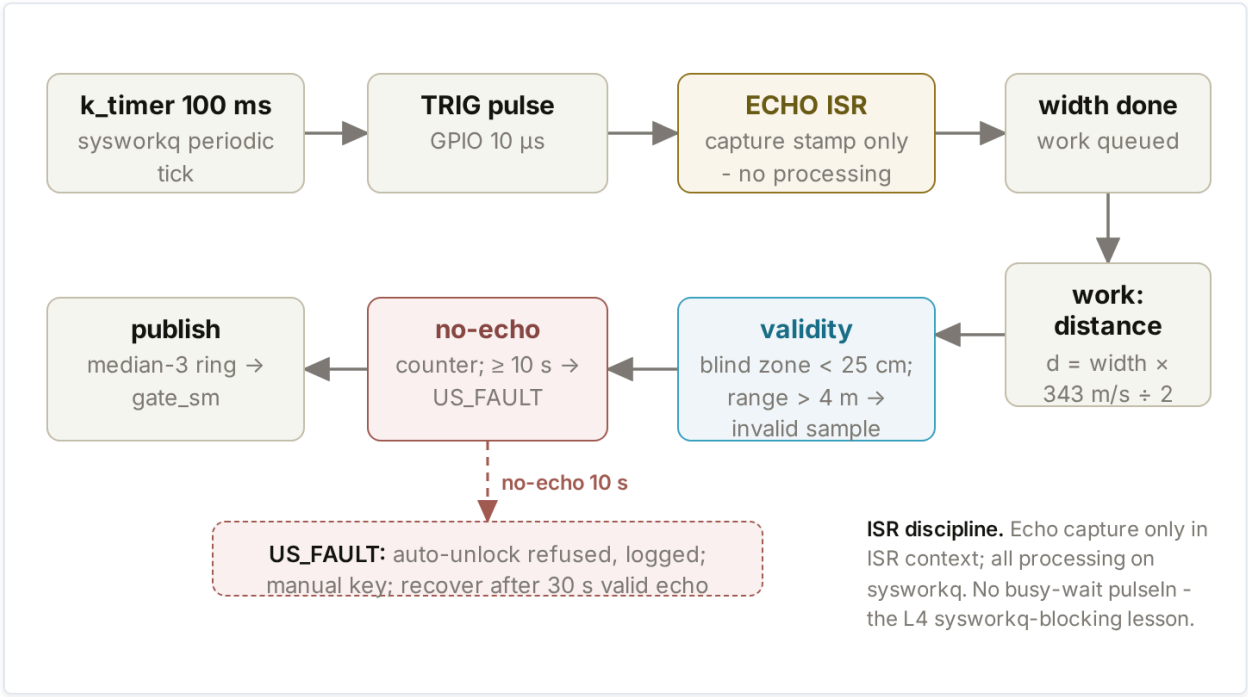


Figure 9: Ultrasonic capture discipline: hardware PPI/TIMER captures timestamps; median-of-3 and temperature compensation execute on sysworkq.

16 Task Allocation and Scheduling

Execution Context	Responsible Operations	Execution Constraints & Rules
Hardware PPI & Timers	ECHO rising and falling edge timestamp capture.	Zero software latency; completely independent of CPU execution and BLE radio interrupts.
GPIO ISR Context	Reed pin state change interrupt.	Queues delayed work to `sysworkq`; no printk, no mutex locks, bounded to <5 µs.
System Workqueue (sysworkq)	Ultrasonic 100 ms trigger work item, median-of-3 filter, temperature compensation, reed debouncing, `gate_sm` transitions, relay control, NVS commits.	Non-blocking; all work items <5 ms; no `k_sleep`. Advances cooperative WDT progress tick. Stack: 2048 bytes.
Main Thread	Sequential subsystem initialization, CDC ACM shell command processing, cooperative WDT watchdog feeding loop.	Runs at lowest priority; feeds hardware WDT only when `sysworkq` progress tick has incremented. Stack: 1024 bytes.
Zephyr Bluetooth Threads	Link-layer Controller, HCI driver, RX thread.	Callbacks strictly offload events to `sysworkq` via message queues. Guarded against flash stalls by `CONFIG_BT_CTLR_FLASH_GUARD=y`.

17 Verification Protocol (T-Series)

ID	Verification Test	Field Procedure Summary	Strict Pass / Fail Criteria
T-A	**Ultrasonic Grid ROC & Blind Zone**	Grid test 0.05 m to 3.0 m in 0.1 m increments. Measure distance accuracy, jitter (sd), and 0.05–0.25 m blind-zone response.	≥ 95% detection within `maxDetectableDistance`; <5% false detections outside hysteresis band; stationary jitter sd <2 cm. Blind-zone behavior verified.
T-B	**Walk-by & Latency Validation**	10 walk-by runs along sidewalk (2 m offset); 10 real arrival runs with smartphone pocketed.	Zero unlocks on walk-bys; end-to-end unlock latency between 2.5–3.5 s; clean trace IDLE → LINK → UNLOCK_HOLD → ARM.
T-C	**Exit Grace & Gate Rebound Injection**	Owner exits and closes gate; injects gate rebound (gate unlatches and bounces shut within 250 ms).	Zero unlocks during standard 60 s exit grace; rebound injection recovers to IDLE within 3 s without triggering 60 s lockout.
T-E	**HIL Fault Injection**	Scripted disconnection of ultrasonic sensor, BLE drop during UNLOCK_HOLD, reed wire severance, forced CPU hang, and power-cycle during BLOCKED.	Every fault terminates in fail-secure locked state; strike energization never exceeds 10 s; lockout state survives power reset; WDT trips on CPU stall.
T-F	**Connection Zone Mapping**	Grid walk inside house (4–5 m) and outside walkway with phone in pocket/hand across sweep of `txPowerDbm` (-8 to +4 dBm).	Reliable connection on exterior approach (≥ 95%); zero connections at interior 4–5 m positions; calibrated power committed to NVS.
T-D	**Weather & Environmental Hardening**	Simulated torrential rain, icing on transducer face, and enclosure thermal cycling.	Deferred to Hardening Phase 2.

18 Open Items and Hardening Backlog

Backlog Item	Classification	Re-entry Phase & Implementation Notes
Actuator Flyback & Power Decoupling	Hardware Hardening (Phase 2)	Install 1N4007 + 15 V TVS across strike terminals and 1000 µF buffer on MCU rail before final production field deployment.
Ultrasonic Sensor Production Migration	Hardware Hardening (Phase 2)	Transition from bench HC-SR04 to weather-sealed JSN-SR04T mounted in 20 cm recessed housing with acoustic hood.
Continuous Strike Duty Confirmation	Bench Data Check	Verify strike solenoid resistance and thermal dissipation limit; confirm `failedAttemptTimeoutMs` ≤ rated max-on.
Tamper & Latch-Monitoring Strike	Hardening Phase 2	Incorporate integrated microswitch electric strike to monitor true bolt engagement and enclosure tamper switch.
Mobile Companion Application	Software Phase 6	Deploy Android foreground service and iOS CoreBluetooth background connection manager; implement user lockout notification UX.

Appendix A: Parameter Reference

Normative master parameter reference. All values default from firmware headers, persist in NVS with CRC32, and are bounds-checked at runtime.

Parameter Key Name	Default	Permitted Range	Calibration Method & Constraint Rationale
`maxDetectableDistance_cm`	150	60 – 250	Calibrated via T-A ROC. Set above blind zone + approach walkway margin.
`usHysteresis_cm`	10	5 – 25	Prevents boundary chatter. Arm at distance + band; disarm at distance - band.
`usUnlockDwellMs`	2000	1000 – 3000	Balances hands-free responsiveness against walk-by filtering (Race 4).
`usDebounceN_M`	3-of-5	2-of-3 to 5-of-8	Window open uses fast 2-of-3; unlock decision evaluation uses strict 3-of-5.
`exitGraceMs`	60000	30000 – 120000	Covers exit gate closure, walk-away, and phone reconnection (Race 2).
`reconnectHoldoffMs`	10000	5000 – 30000	BLE disconnect cooldown after an exit event while reed is OPEN.
`failedAttemptTimeoutMs`	10000	3000 – 15000	Maximum strike energization time. MUST NOT exceed strike rated continuous on-time.
`strikeCooldownMs`	10000	10000 – 60000	Enforces ≤ 50% duty cycle before allowing any unlock re-attempt.
`failedAttemptCounter`	3	2 – 5	Consecutive unlatched attempt timeouts before locking out user.
`blockDurationMin`	15	5 – 60	Lockout duration for blocked user. Manual key remains functional.
`usFaultThresholdMs`	10000	5000 – 30000	Duration of missing or blind echo required to declare `US_FAULT`.
`txPowerDbm`	0	-8 to +4	Calibrated via T-F grid walk. Trims connection bubble to approach walkway.
`wdtTimeoutS`	8	4 – 16	Hardware watchdog reset period; accounts for worst-case workqueue latency.
`pairingWindowS`	60	30 – 120	Administrative pairing window duration. Zero outside this window.
`smpFailTimeoutMs`	2500	1500 – 4000	Security handshake timeout; frees single BLE slot quickly on rogue connection.
`reedDebounceOpenMs`	30	10 – 50	Break debounce. Instant response to human gate push (<300 ms); re-arms strike.
`reedDebounceCloseMs`	150	50 – 300	Make debounce. Absorbs heavy metal gate rebound chatter against latch stop.
`bleAdvFastIntervalMinMs`	20	20 – 30	Fast BLE advertising interval to catch iOS/Android background scan windows.
`bleAdvFastIntervalMaxMs`	30	25 – 40	Upper bound of fast advertising window during US presence.
`tempCompEnabled`	true	true / false	Enables nRF52840 internal `TEMP` acoustic velocity compensation.

Appendix B: Requirements Traceability Matrix

Story ID	Functional Requirements	Design Chapter References	Verification Protocol
US-1 (Arrival)	FR-1, FR-2, FR-3	Chapters 9, 10; Figure 5	T-B, T-F
US-2 (Exit + Close)	FR-4, FR-5, FR-6	Chapter 10; Figure 6	T-C
US-3 (Manual Key)	FR-13	Chapter 6; Table 10	T-E
US-4 (Blocked Feedback)	FR-8, FR-9, FR-10	Chapters 12, 13; Figure 7	T-E
US-5 (Calibration)	FR-9	Chapter 9; Appendix A	T-A, T-F
US-6 (No Spurious Unlock)	FR-2, FR-5	Chapter 9; Table 14	T-B, T-C, T-F
US-7 (Power Outage)	FR-9, FR-12	Chapters 6, 10	T-E
US-8 (Stranger Denial)	FR-1, FR-2	Chapters 9, 12	T-B, T-F
US-9 (Observability)	FR-14	Chapters 14, 16	All T-Series Logs
M-1 (Unbonded Phone)	FR-1	Chapter 12	T-F
M-2 (Accomplice Race 1)	FR-2	Chapters 11, 12	T-F
M-3 (Gate-Shaker)	FR-7, FR-8	Chapter 10; Figure 7	T-E
M-4 (Blinder DoS)	FR-11	Chapters 10, 13	T-E
M-5 (Pairing Fraud)	FR-15	Chapter 12	T-E

Appendix C: Glossary and References

Term / Abbreviation	Technical Definition
Fail-Secure Strike	An electromechanical lock mechanism that remains mechanically locked in the absence of electrical power. Power is required to unlatch.
Auto-Latch	A mechanical latch that automatically throws into the strike keeper when the gate leaf closes, physically locking the gate without electrical actuation.
LE Secure Connections	Bluetooth 4.2+ cryptographic pairing mechanism utilizing Elliptic Curve Diffie-Hellman (ECDH) P-256 with authenticated passkey entry.
PPI (Programmable Peripheral Interconnect)	Nordic Semiconductor on-chip crossbar interconnect enabling hardware peripherals to communicate autonomously without CPU or RTOS interrupt latency.
US-Gated Connection Window	Operational discipline wherein the gate's BLE radio transmitter remains completely off/silent until physical human presence is confirmed by ultrasonic reflection.
exitGrace	A timed lockout period (60 s) initiated by the reed switch closing, during which automatic unlocking is hard-refused to prevent accidental opening behind an exiting resident.
sysworkq	The Zephyr RTOS system workqueue; the primary cooperative thread responsible for running event-driven sensor processing, state machine transitions, and relay driving.

C.1 Standards & Field References

- T0/T1/T2 On-Field Measurement Dataset & L4-A Analysis Report, October 2026.
- Bluetooth Core Specification v5.4, Volume 3 (Host), Part H (Security Manager Specification).
- Nordic Semiconductor nRF52840 Product Specification v1.7 (NVMC, PPI, TIMER, TEMP).
- Zephyr RTOS Project Documentation v4.4 (Workqueue discipline, Bluetooth Controller Flash Guard).